Partial Class
&
Partial Method

Partial Class and Partial Method

- Each class in C# resides in a separate physical file with a .cs extension. C# provides the ability to have a single class implementation in multiple .cs files using the ***partial*** modifier keyword. The *partial* modifier can be applied to a class, method, interface or structure.

- A partial class or struct may contain partial methods. A partial method must be declared in one of the partial classes. A partial method may or may not have an implementation. If the partial method doesn't have an implementation in any part then the compiler will not generate that method in the final class.

```
public partial class MyPartialClass{
    public MyPartialClass(){
    }

    //partial method. implicitly private
    partial void PartialMethod(int val);

    public void Method1(int val){
        Console.WriteLine(val);
    }
}
```

PartialClassFile1.cs

```
public partial class MyPartialClass{

    public void Method2(int val){
        Console.WriteLine(val);
    }

    partial void PartialMethod(int val){
        Console.WriteLine(val);
    }
}
```

PartialClassFile2.cs

Compiles as a single class

```
public class MyPartialClass{
    public MyPartialClass(){
    }

    private void PartialMethod(int val){
        Console.WriteLine(val);
    }

    public void Method1(int val){
        Console.WriteLine(val);
    }

    public void Method2(int val){
        Console.WriteLine(val);
    }
}
```

Rules for partial class and method

- All the partial class definitions must be in the same assembly and namespace.
- All the parts must have the same accessibility like public or private, etc.
- If any part is declared abstract, sealed or base type then the whole class is declared of the same type.
- Different parts can have different base types and so the final class will inherit all the base types.
- The Partial modifier can only appear immediately before the keywords class, struct, or interface.
- Nested partial types are allowed.
- The partial method declaration must began with the partial modifier.
- The partial method can have a ref but not an out parameter.
- Partial methods are implicitly private methods.
- Partial methods can be static methods.
- Partial methods can be generic.

Advantages of Partial class

- Multiple developers can work simultaneously with a single class in separate files.
- When working with automatically generated source, code can be added to the class without having to recreate the source file. For example, Visual Studio separates HTML code for the UI and server side code into two separate files: .aspx and .cs files.